

Civil Engineering

Algebra and Coordinate Geometry

Professional Learner Notes

Indices, logarithms, equations, transposition, simultaneous equations and coordinate geometry for engineering layouts.

Section	Purpose
Indices and standard form	Engineering notation, powers and scientific-unit calculations.
Logarithms	Laws of logarithms and reduction of laws to linear form.
Equations and formula transposition	Solving unknowns in engineering relationships.
Simultaneous equations	Solving two-variable site and quantity problems.
Coordinate geometry	Distance, midpoint, gradient and straight-line equations.

How to Study This Topic

The learning method follows a programmed engineering mathematics approach: understand the concept, study the formula, follow a worked method, then practise with a technical problem. The aim is not memorisation only; the learner must interpret the answer in the department context.

- Write the formula before substitution.
- Use correct units.
- Check whether the result is reasonable.
- State a technical interpretation.
- Repeat practice until the method is fluent.

Formula Summary

Index multiplication

$$a^m a^n = a^{m+n}$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Index division

$$\frac{a^m}{a^n} = a^{m-n}$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Product law of logarithms

$$\log(ab) = \log a + \log b$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Distance formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Line equation

$$y = mx + c$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

1. Indices and standard form

Engineering notation, powers and scientific-unit calculations.

In civil engineering, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$a^m a^n = a^{m+n}$$

Worked method

Example: Control-point distance. Two survey control points have coordinates A(120, 250) and B(180, 330). Find AB.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values. Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports control points. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving setting out. Show formula, substitution, units and interpretation.
2. Create and solve one technical calculation involving material equations. Show formula, substitution, units and interpretation.
3. Create and solve one technical calculation involving linear models. Show formula, substitution, units and interpretation.
4. Create and solve one technical calculation involving control points. Show formula, substitution, units and interpretation.
5. Create and solve one technical calculation involving road profiles. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving setting out. Show formula, substitution, units and interpretation.
7. Create and solve one technical calculation involving material equations. Show formula, substitution, units and interpretation.
8. Create and solve one technical calculation involving linear models. Show formula, substitution, units and interpretation.

2. Logarithms

Laws of logarithms and reduction of laws to linear form.

In civil engineering, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$\frac{a^m}{a^n} = a^{m-n}$$

Worked method

Example: Road gradient line. A road profile is approximated by $y = 0.04x + 102$. Find the level at $x = 75$ m.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values. Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports road profiles. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving material equations. Show formula, substitution, units and interpretation.
2. Create and solve one technical calculation involving linear models. Show formula, substitution, units and interpretation.
3. Create and solve one technical calculation involving control points. Show formula, substitution, units and interpretation.
4. Create and solve one technical calculation involving road profiles. Show formula, substitution, units and interpretation.
5. Create and solve one technical calculation involving setting out. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving material equations. Show formula, substitution, units and interpretation.
7. Create and solve one technical calculation involving linear models. Show formula, substitution, units and interpretation.
8. Create and solve one technical calculation involving control points. Show formula, substitution, units and interpretation.

3. Equations and formula transposition

Solving unknowns in engineering relationships.

In civil engineering, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$\log(ab) = \log a + \log b$$

Worked method

Example: Material equation. Solve simultaneous equations representing cement and sand quantities in two mortar mixes.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values. Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports setting out. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving linear models. Show formula, substitution, units and interpretation.
2. Create and solve one technical calculation involving control points. Show formula, substitution, units and interpretation.
3. Create and solve one technical calculation involving road profiles. Show formula, substitution, units and interpretation.
4. Create and solve one technical calculation involving setting out. Show formula, substitution, units and interpretation.
5. Create and solve one technical calculation involving material equations. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving linear models. Show formula, substitution, units and interpretation.
7. Create and solve one technical calculation involving control points. Show formula, substitution, units and interpretation.
8. Create and solve one technical calculation involving road profiles. Show formula, substitution, units and interpretation.

4. Simultaneous equations

Solving two-variable site and quantity problems.

In civil engineering, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Worked method

Example: Control-point distance. Two survey control points have coordinates A(120, 250) and B(180, 330). Find AB.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values. Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports material equations. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving control points. Show formula, substitution, units and interpretation.
2. Create and solve one technical calculation involving road profiles. Show formula, substitution, units and interpretation.
3. Create and solve one technical calculation involving setting out. Show formula, substitution, units and interpretation.
4. Create and solve one technical calculation involving material equations. Show formula, substitution, units and interpretation.
5. Create and solve one technical calculation involving linear models. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving control points. Show formula, substitution, units and interpretation.
7. Create and solve one technical calculation involving road profiles. Show formula, substitution, units and interpretation.
8. Create and solve one technical calculation involving setting out. Show formula, substitution, units and interpretation.

5. Coordinate geometry

Distance, midpoint, gradient and straight-line equations.

In civil engineering, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$y=mx+c$$

Worked method

Example: Road gradient line. A road profile is approximated by $y = 0.04x + 102$. Find the level at $x = 75$ m.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values. Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports linear models. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving road profiles. Show formula, substitution, units and interpretation.
2. Create and solve one technical calculation involving setting out. Show formula, substitution, units and interpretation.
3. Create and solve one technical calculation involving material equations. Show formula, substitution, units and interpretation.
4. Create and solve one technical calculation involving linear models. Show formula, substitution, units and interpretation.
5. Create and solve one technical calculation involving control points. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving road profiles. Show formula, substitution, units and interpretation.
7. Create and solve one technical calculation involving setting out. Show formula, substitution, units and interpretation.
8. Create and solve one technical calculation involving material equations. Show formula, substitution, units and interpretation.

Mixed Revision Questions

1. A civil engineering problem involves road profiles. Choose a suitable mathematical method, solve the problem, and explain the result.
2. A civil engineering problem involves setting out. Choose a suitable mathematical method, solve the problem, and explain the result.
3. A civil engineering problem involves material equations. Choose a suitable mathematical method, solve the problem, and explain the result.
4. A civil engineering problem involves linear models. Choose a suitable mathematical method, solve the problem, and explain the result.
5. A civil engineering problem involves control points. Choose a suitable mathematical method, solve the problem, and explain the result.
6. A civil engineering problem involves road profiles. Choose a suitable mathematical method, solve the problem, and explain the result.
7. A civil engineering problem involves setting out. Choose a suitable mathematical method, solve the problem, and explain the result.
8. A civil engineering problem involves material equations. Choose a suitable mathematical method, solve the problem, and explain the result.
9. A civil engineering problem involves linear models. Choose a suitable mathematical method, solve the problem, and explain the result.
10. A civil engineering problem involves control points. Choose a suitable mathematical method, solve the problem, and explain the result.
11. A civil engineering problem involves road profiles. Choose a suitable mathematical method, solve the problem, and explain the result.
12. A civil engineering problem involves setting out. Choose a suitable mathematical method, solve the problem, and explain the result.
13. A civil engineering problem involves material equations. Choose a suitable mathematical method, solve the problem, and explain the result.
14. A civil engineering problem involves linear models. Choose a suitable mathematical method, solve the problem, and explain the result.
15. A civil engineering problem involves control points. Choose a suitable mathematical method, solve the problem, and explain the result.
16. A civil engineering problem involves road profiles. Choose a suitable mathematical method, solve the problem, and explain the result.
17. A civil engineering problem involves setting out. Choose a suitable mathematical method, solve the problem, and explain the result.
18. A civil engineering problem involves material equations. Choose a suitable mathematical method, solve the problem, and explain the result.
19. A civil engineering problem involves linear models. Choose a suitable mathematical method, solve the problem, and explain the result.
20. A civil engineering problem involves control points. Choose a suitable mathematical method, solve the problem, and explain the result.
21. A civil engineering problem involves road profiles. Choose a suitable mathematical method, solve the problem, and explain the result.

22. A civil engineering problem involves setting out. Choose a suitable mathematical method, solve the problem, and explain the result.
23. A civil engineering problem involves material equations. Choose a suitable mathematical method, solve the problem, and explain the result.
24. A civil engineering problem involves linear models. Choose a suitable mathematical method, solve the problem, and explain the result.
25. A civil engineering problem involves control points. Choose a suitable mathematical method, solve the problem, and explain the result.
26. A civil engineering problem involves road profiles. Choose a suitable mathematical method, solve the problem, and explain the result.
27. A civil engineering problem involves setting out. Choose a suitable mathematical method, solve the problem, and explain the result.
28. A civil engineering problem involves material equations. Choose a suitable mathematical method, solve the problem, and explain the result.
29. A civil engineering problem involves linear models. Choose a suitable mathematical method, solve the problem, and explain the result.
30. A civil engineering problem involves control points. Choose a suitable mathematical method, solve the problem, and explain the result.

References and Learning Sources

These learner materials are original notes prepared for Mr Mwirigi Mathematics Training Hub. The topic sequencing and level are informed by standard engineering mathematics references and the uploaded curriculum/OS materials. No textbook pages or solution-manual pages have been reproduced.

1. John Bird, Engineering Mathematics, Newnes / Elsevier.
2. K. A. Stroud and Dexter J. Booth, Engineering Mathematics, Palgrave Macmillan.
3. Uploaded KNEC solved mathematics materials for examination-style orientation.
4. Civil Engineering, Building Technology and Land Survey curriculum and occupational-standard materials used for local alignment.